Ch 3.6 R Lab

In [2]:
```
library(MASS)
```

In [3]:
```
library(ISLR)
```

In [4]:
```
names(Boston)
```

'crim' · 'zn' · 'indus' · 'chas' · 'nox' · 'rm' · 'age' · 'dis' · 'rad' · 'tax' · 'ptratio' · 'black' · 'lstat' · 'medv'

In [5]:
```
lm.fit=lm(medv~lstat,data=Boston)
```

In [6]:
```
attach(Boston)
```

In [7]:
```
lm.fit=lm(medv~lstat)
```

In [8]:
```
lm.fit
```

```
Call:
lm(formula = medv ~ lstat)

Coefficients:
(Intercept)        lstat
      34.55        -0.95
```

In [9]:
```
summary(lm.fit)
```

```
Call:
lm(formula = medv ~ lstat)

Residuals:
    Min      1Q  Median      3Q     Max
-15.168  -3.990  -1.318   2.034  24.500

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 34.55384    0.56263   61.41   <2e-16 ***
lstat       -0.95005    0.03873  -24.53   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.216 on 504 degrees of freedom
Multiple R-squared:  0.5441,    Adjusted R-squared:  0.5432
F-statistic: 601.6 on 1 and 504 DF,  p-value: < 2.2e-16
```

In [10]:
```
names(lm.fit)
```

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'coefficients' · 'residuals' · 'effects' · 'rank' · 'fitted.values' · 'assign' · 'qr' · 'df.residual' · 'xlevels' · 'call' · 'terms' · 'model'

In [11]:
```
coef(lm.fit)
```

(Intercept): 34.5538408793831 **lstat:** -0.95004935375799

In [12]:
```
confint(lm.fit)
```

A matrix: 2 × 2 of type dbl

	2.5 %	97.5 %
(Intercept)	33.448457	35.6592247
lstat	-1.026148	-0.8739505

In [13]:
```
predict(lm.fit,data.frame(lstat=(c(5,10,15))),interval="confidence")
```

A matrix: 3 × 3 of type dbl

	fit	lwr	upr
1	29.80359	29.00741	30.59978
2	25.05335	24.47413	25.63256
3	20.30310	19.73159	20.87461

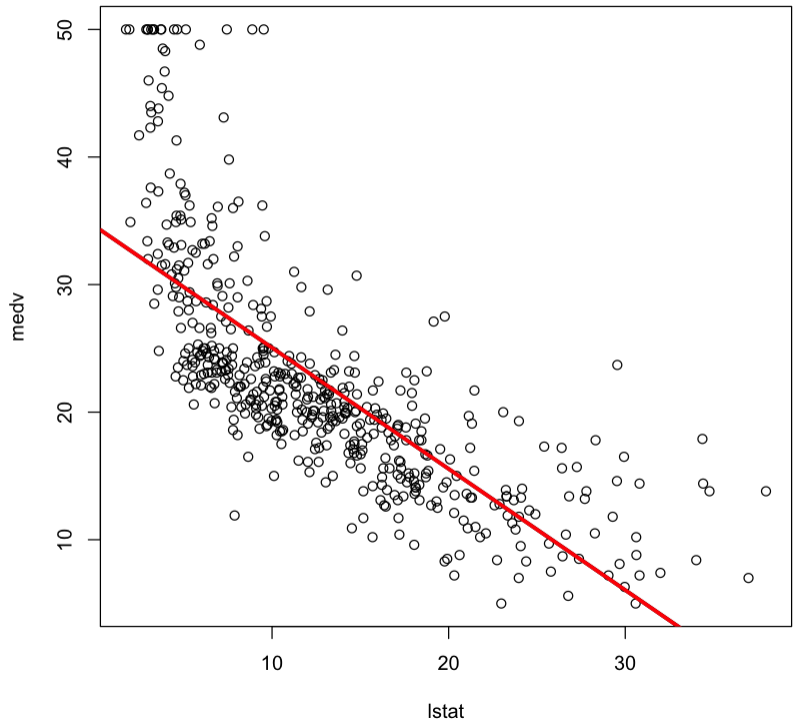
In [14]:
```
predict(lm.fit,data.frame(lstat=(c(5,10,15))),interval="prediction")
```

A matrix: 3 × 3 of type dbl

	fit	lwr	upr
1	29.80359	17.565675	42.04151
2	25.05335	12.827626	37.27907
3	20.30310	8.077742	32.52846

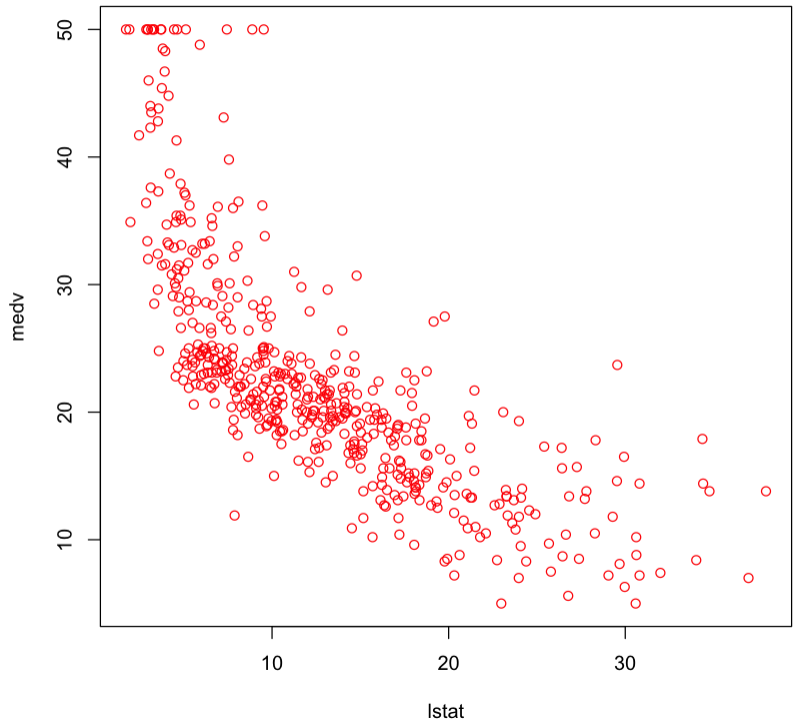
In [15]:
```
plot(lstat,medv)
abline(lm.fit)
abline(lm.fit,lwd=3)
abline(lm.fit,lwd=3,col="red")
```

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In [16]:

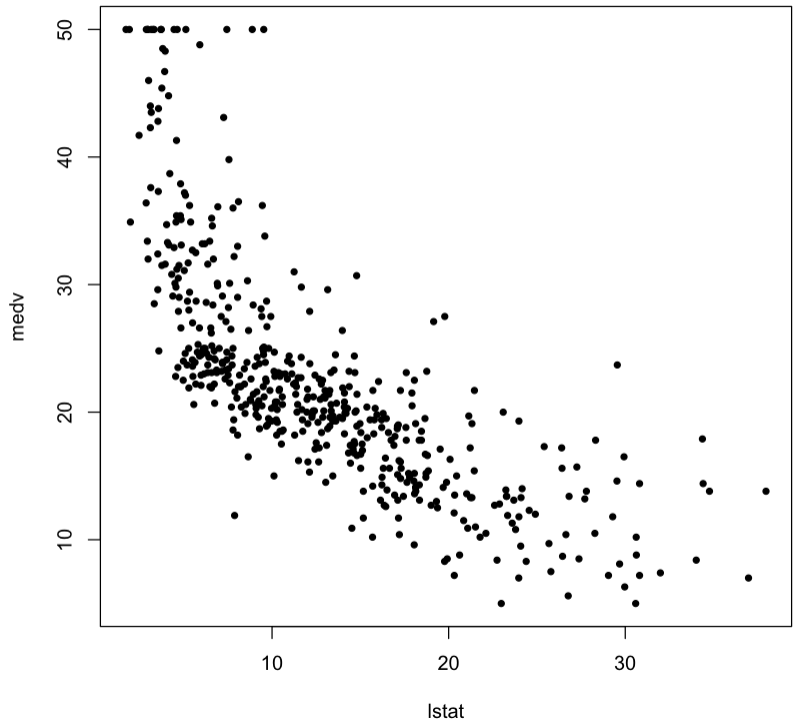
```
plot(lstat,medv,col="red")
```



In [17]:

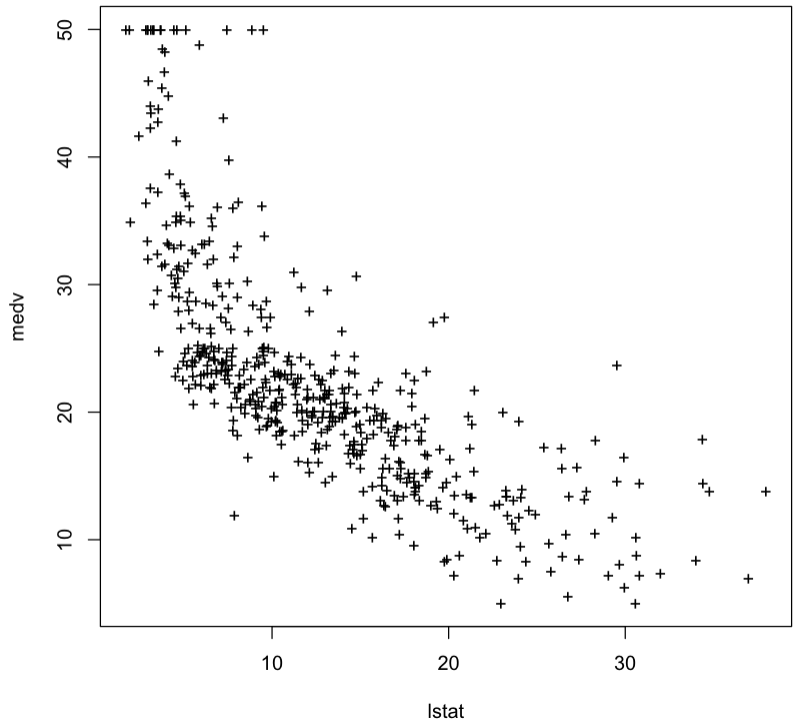
```
plot(lstat,medv,pch=20)
```

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In [18]:

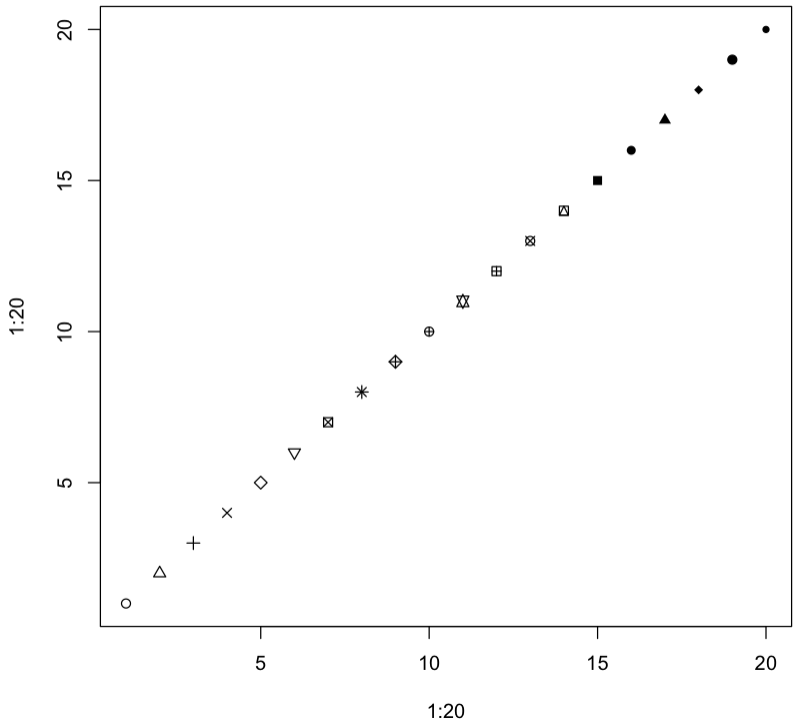
```
plot(lstat,medv,pch="+")
```



In [19]:

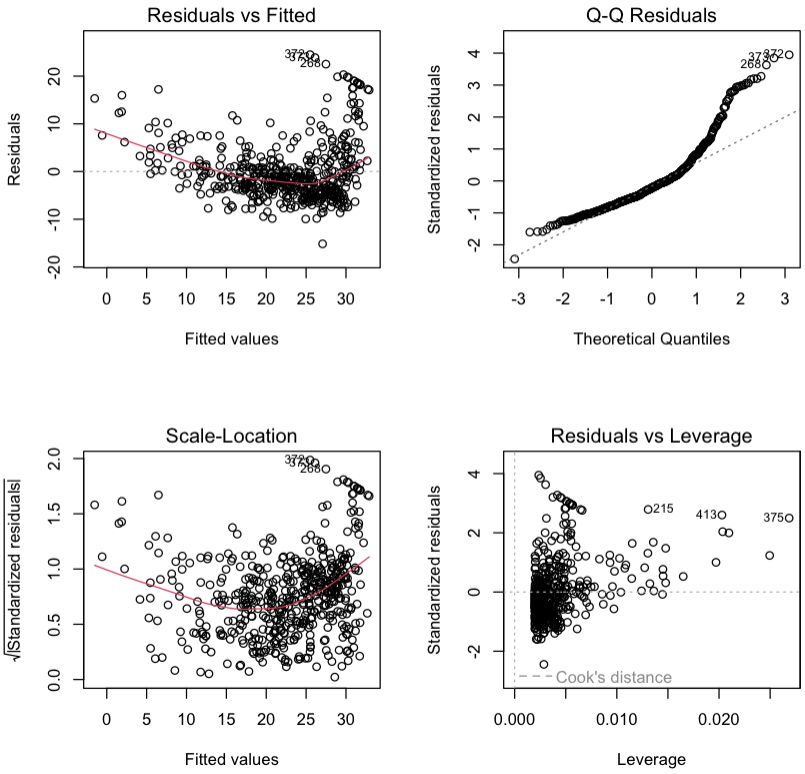
```
plot(1:20,1:20,pch=1:20)
```

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In [20]:

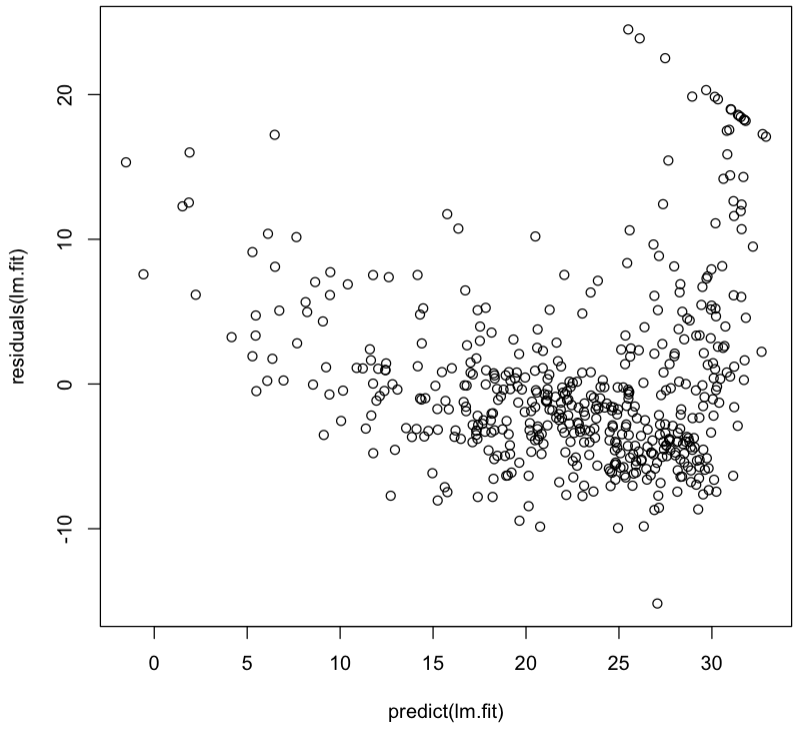
```
par(mfrow=c(2,2))
plot(lm.fit)
```



In [21]:

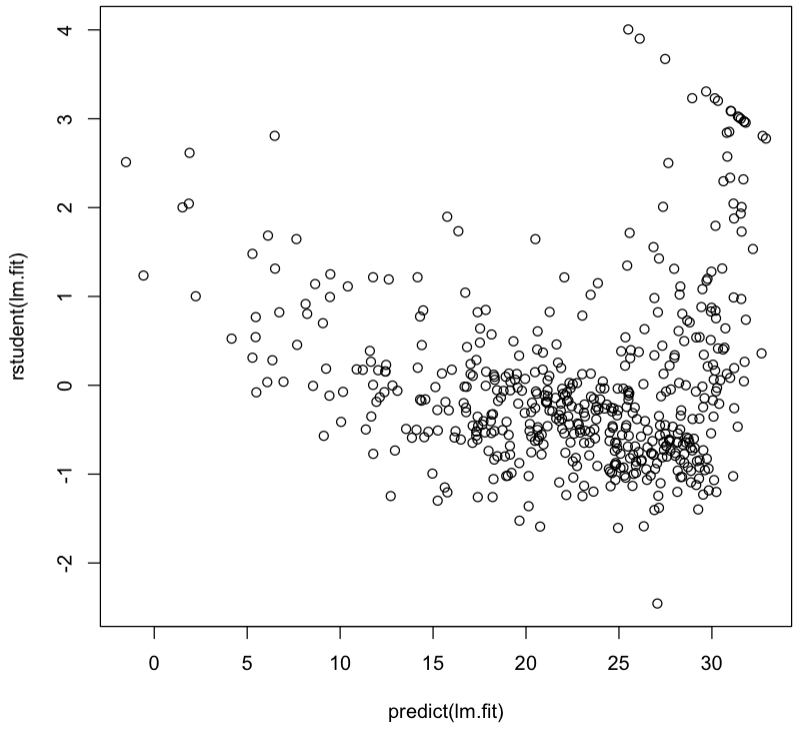
```
plot(predict(lm.fit),residuals(lm.fit))
```

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In [22]:

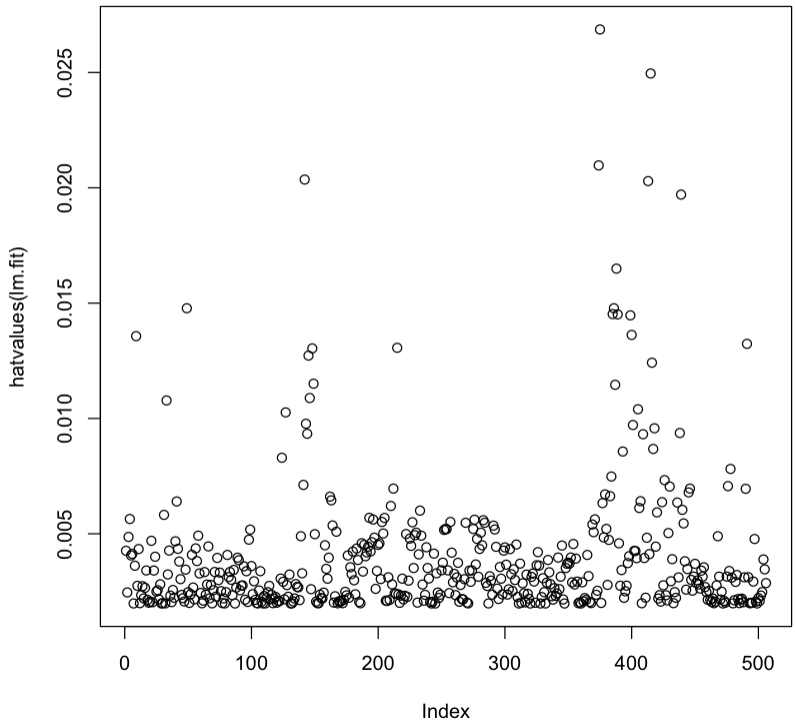
```
plot(predict(lm.fit),rstudent(lm.fit))
```



In [23]:

```
plot(hatvalues(lm.fit))
```

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In [24]:

```
which.max(hatvalues(lm.fit))
```

375: 375

In [25]:

```
lm.fit=lm(medv~lstat+age,data=Boston)
```

In [26]:

```
summary(lm.fit)
```

```
Call:
lm(formula = medv ~ lstat + age, data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max 
-15.981  -3.978  -1.283   1.968  23.158 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 33.22276    0.73085  45.458  < 2e-16 ***
lstat       -1.03207    0.04819 -21.416  < 2e-16 ***
age          0.03454    0.01223   2.826  0.00491 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.173 on 503 degrees of freedom
Multiple R-squared:  0.5513,    Adjusted R-squared:  0.5495 
F-statistic:   309 on 2 and 503 DF,  p-value: < 2.2e-16
```

In [27]:

```
lm.fit=lm(medv~.,data=Boston)
```

In [28]:

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```
summary(lm.fit)
```

```
Call:
lm(formula = medv ~ ., data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max
-15.595  -2.730  -0.518   1.777  26.199

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.646e+01  5.103e+00   7.144 3.28e-12 ***
crim        -1.080e-01  3.286e-02  -3.287 0.001087 **
zn           4.642e-02  1.373e-02   3.382 0.000778 ***
indus        2.056e-02  6.150e-02   0.334 0.738288
chas         2.687e+00  8.616e-01   3.118 0.001925 **
nox         -1.777e+01  3.820e+00  -4.651 4.25e-06 ***
rm           3.810e+00  4.179e-01   9.116  < 2e-16 ***
age          6.922e-04  1.321e-02   0.052 0.958229
dis         -1.476e+00  1.995e-01  -7.398 6.01e-13 ***
rad          3.060e-01  6.635e-02   4.613 5.07e-06 ***
tax         -1.233e-02  3.760e-03  -3.280 0.001112 **
ptratio     -9.527e-01  1.308e-01  -7.283 1.31e-12 ***
black        9.312e-03  2.686e-03   3.467 0.000573 ***
lstat       -5.248e-01  5.072e-02 -10.347  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.745 on 492 degrees of freedom
Multiple R-squared:  0.7406,    Adjusted R-squared:  0.7338
F-statistic: 108.1 on 13 and 492 DF,  p-value: < 2.2e-16
```

In [29]:

```
library(car)
```

```
Loading required package: carData
```

In [30]:

```
vif(lm.fit)
```

crim: 1.79219154743325 **zn:** 2.29875817874944 **indus:** 3.99159641834604 **chas:** 1.07399532755379 **nox:** 4.39371984757748 **rm:** 1.93374443578326 **age:** 3.10082551281534 **dis:** 3.95594490637273 **rad:** 7.48449633527447 **tax:** 9.00855394759708 **ptratio:** 1.7990840492489 **black:** 1.34852107640638 **lstat:** 2.94149107809194

In [31]:

```
lm.fit1=lm(medv~.-age,data=Boston)
```

In [32]:

```
summary(lm.fit1)
```

```
Call:
lm(formula = medv ~ . - age, data = Boston)

Residuals:
     Min       1Q   Median       3Q      Max
-15.6054  -2.7313  -0.5188   1.7601  26.2243
```

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```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  36.436927   5.080119   7.172 2.72e-12 ***
crim         -0.108006   0.032832  -3.290 0.001075 **
zn            0.046334   0.013613   3.404 0.000719 ***
indus         0.020562   0.061433   0.335 0.737989
chas          2.689026   0.859598   3.128 0.001863 **
nox         -17.713540   3.679308  -4.814 1.97e-06 ***
rm            3.814394   0.408480   9.338  < 2e-16 ***
dis          -1.478612   0.190611  -7.757 5.03e-14 ***
rad           0.305786   0.066089   4.627 4.75e-06 ***
tax          -0.012329   0.003755  -3.283 0.001099 **
ptratio      -0.952211   0.130294  -7.308 1.10e-12 ***
black         0.009321   0.002678   3.481 0.000544 ***
lstat        -0.523852   0.047625 -10.999  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.74 on 493 degrees of freedom
Multiple R-squared:  0.7406,    Adjusted R-squared:  0.7343
F-statistic: 117.3 on 12 and 493 DF,  p-value: < 2.2e-16
```

In [33]:

```
lm.fit1=update(lm.fit,~.-age)
```

In [34]:

```
summary(lm(medv~lstat*age,data=Boston))
```

```
Call:
lm(formula = medv ~ lstat * age, data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max
-15.806  -4.045  -1.333   2.085  27.552

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 36.0885359  1.4698355  24.553  < 2e-16 ***
lstat       -1.3921168  0.1674555  -8.313 8.78e-16 ***
age         -0.0007209  0.0198792  -0.036   0.9711
lstat:age    0.0041560  0.0018518   2.244   0.0252 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.149 on 502 degrees of freedom
Multiple R-squared:  0.5557,    Adjusted R-squared:  0.5531
F-statistic: 209.3 on 3 and 502 DF,  p-value: < 2.2e-16
```

In [35]:

```
lm.fit2=lm(medv~lstat+I(lstat^2))
```

In [36]:

```
summary(lm.fit2)
```

```
Call:
lm(formula = medv ~ lstat + I(lstat^2))

Residuals:
     Min       1Q   Median       3Q      Max
-15.2834  -3.8313  -0.5295   2.3095  25.4148
```

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```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 42.862007   0.872084   49.15   <2e-16 ***
lstat       -2.332821   0.123803  -18.84   <2e-16 ***
I(lstat^2)   0.043547   0.003745   11.63   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.524 on 503 degrees of freedom
Multiple R-squared:  0.6407,    Adjusted R-squared:  0.6393
F-statistic: 448.5 on 2 and 503 DF,  p-value: < 2.2e-16
```

In [37]:

```
lm.fit=lm(medv~lstat)
```

In [38]:

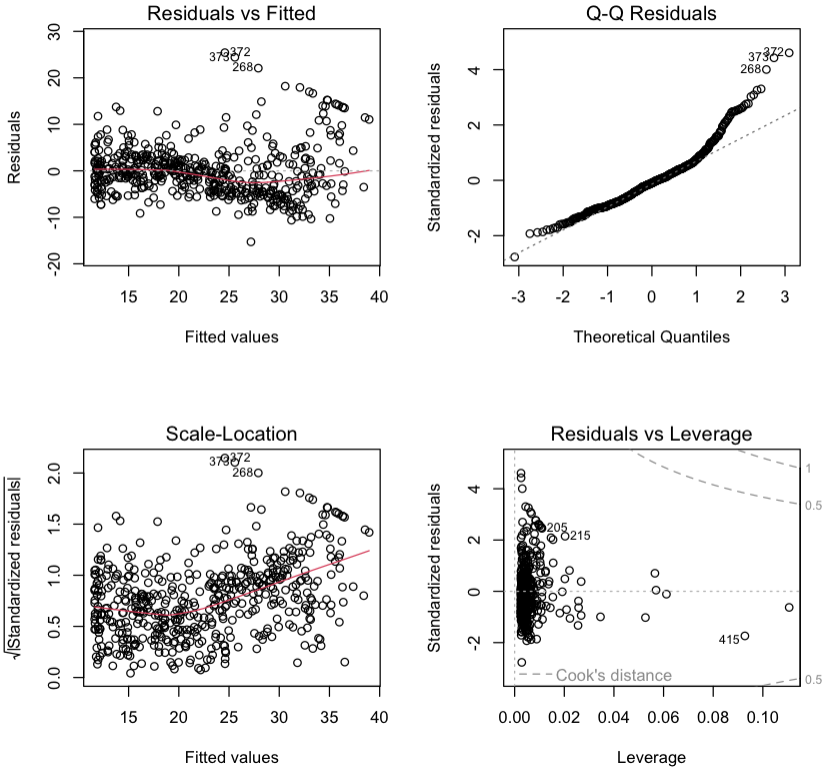
```
anova(lm.fit,lm.fit2)
```

A anova: 2 × 6

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	504	19472.38	NA	NA	NA	NA
2	503	15347.24	1	4125.138	135.1998	7.630116e-28

In [39]:

```
par(mfrow=c(2,2))
plot(lm.fit2)
```



In [40]:

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```
lm.fit5=lm(medv~poly(lstat,5))
```

In [41]:

```
summary(lm.fit5)
```

```
Call:
lm(formula = medv ~ poly(lstat, 5))

Residuals:
     Min       1Q   Median       3Q      Max 
-13.5433  -3.1039  -0.7052   2.0844  27.1153 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)       22.5328     0.2318  97.197  < 2e-16 ***
poly(lstat, 5)1 -152.4595     5.2148 -29.236  < 2e-16 ***
poly(lstat, 5)2   64.2272     5.2148  12.316  < 2e-16 ***
poly(lstat, 5)3  -27.0511     5.2148  -5.187 3.10e-07 ***
poly(lstat, 5)4   25.4517     5.2148   4.881 1.42e-06 ***
poly(lstat, 5)5  -19.2524     5.2148  -3.692 0.000247 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.215 on 500 degrees of freedom
Multiple R-squared:  0.6817,    Adjusted R-squared:  0.6785 
F-statistic: 214.2 on 5 and 500 DF,  p-value: < 2.2e-16
```

In [42]:

```
summary(lm(medv~log(rm),data=Boston))
```

```
Call:
lm(formula = medv ~ log(rm), data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max 
-19.487  -2.875  -0.104   2.837  39.816 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -76.488      5.028  -15.21   <2e-16 ***
log(rm)       54.055      2.739   19.73   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.915 on 504 degrees of freedom
Multiple R-squared:  0.4358,    Adjusted R-squared:  0.4347 
F-statistic: 389.3 on 1 and 504 DF,  p-value: < 2.2e-16
```

In [43]:

```
names(Carseats)
```

'Sales' · 'CompPrice' · 'Income' · 'Advertising' · 'Population' · 'Price' · 'ShelveLoc' · 'Age' · 'Education' · 'Urban' · 'US'

In [44]:

```
lm.fit=lm(Sales~.+Income:Advertising+Price:Age,data=Carseats)
```

In [45]:

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```
summary(lm.fit)
```

```
Call:
lm(formula = Sales ~ . + Income:Advertising + Price:Age, data = Carseats)

Residuals:
    Min      1Q  Median      3Q     Max
-2.9208 -0.7503  0.0177  0.6754  3.3413

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)         6.5755654  1.0087470   6.519 2.22e-10 ***
CompPrice           0.0929371  0.0041183  22.567  < 2e-16 ***
Income              0.0108940  0.0026044   4.183 3.57e-05 ***
Advertising         0.0702462  0.0226091   3.107 0.002030 **
Population          0.0001592  0.0003679   0.433 0.665330
Price              -0.1008064  0.0074399 -13.549  < 2e-16 ***
ShelveLocGood       4.8486762  0.1528378  31.724  < 2e-16 ***
ShelveLocMedium     1.9532620  0.1257682  15.531  < 2e-16 ***
Age                -0.0579466  0.0159506  -3.633 0.000318 ***
Education          -0.0208525  0.0196131  -1.063 0.288361
UrbanYes            0.1401597  0.1124019   1.247 0.213171
USYes              -0.1575571  0.1489234  -1.058 0.290729
Income:Advertising  0.0007510  0.0002784   2.698 0.007290 **
Price:Age           0.0001068  0.0001333   0.801 0.423812
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.011 on 386 degrees of freedom
Multiple R-squared:  0.8761,    Adjusted R-squared:  0.8719
F-statistic:   210 on 13 and 386 DF,  p-value: < 2.2e-16
```

In [46]:

```
attach(Carseats)
```

In [47]:

```
contrasts(ShelveLoc)
```

A matrix: 3 × 2 of type dbl

	Good	Medium
Bad	0	0
Good	1	0
Medium	0	1

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